

Aritro Saha

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SKILLS

Languages: Python, C++, Go, Java, SQL, TypeScript, JavaScript, Bash / Shell
Cloud & Infrastructure: Google Cloud Platform (GCP), AWS, Docker, Kubernetes, Terraform, Linux / Unix, CI/CD (GitHub Actions)
Frameworks & Big Data: FastAPI, Apache Spark (Databricks), PySpark, DuckDB, LangGraph, Node.js, Next.js, React
Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, DynamoDB, Firestore
Core Engineering: Data Structures & Algorithms, System Design, Distributed Systems, Concurrency, OOP, Asynchronous I/O, REST APIs

WORK EXPERIENCE

Bristol Myers Squibb

Software Engineer 1

Hyderabad, India

Jul 2025 -- Present

- Architected an LLM-driven RAG clinical assistant, cutting diagnostic turnaround time by **30%** for non-small cell lung cancer (NSCLC) cohorts.
- Slashed release overhead by **40+ engineering hours** per cycle by designing a Databricks accelerator with modular microservices and declarative CI/CD.
- Spearheaded end-to-end delivery of a self-service LLM plugin platform adopted across **20+** enterprise teams, replacing manual deployments with event-driven serverless workflows.
- Scaled real-time data ingestion to **5M+ records/hour** with 99.9% availability, deploying asynchronous serverless pipelines via AWS Lambda, Glue, and DynamoDB.

Bajaj Finserv Health

Data Science Engineer Intern

Pune, India

Feb 2025 -- Jun 2025

- Cut claims processing failures by **60%** across **40,000+ daily transactions** by redesigning a multi-worker vision NER extraction pipeline.
- Accelerated processing throughput by **98%** and eliminated external API dependencies by developing a deterministic NER mapping service for ICD-10 codes.
- Reduced P99 API latency by **35% (800ms to 520ms)** by optimizing FastAPI async serialization, database indexing, and regression unit test coverage.

WIPRO

Software Engineering Intern

Kolkata, India

Oct 2023 -- Dec 2023

- Optimized 3D visualization render speeds by **3x** by designing a Three.js engine with procedural geometry, asset compression, and lazy texture loading.
- Boosted deployment frequency by **50%** across staging and production by containerizing microservices with Docker and Azure DevOps CI/CD.

PROJECTS

Structurify: Autonomous ETL Pipeline | GCP, LangGraph, Gemini, DuckDB

GitHub | Live Demo | 2026

- Engineered a serverless fan-out architecture on GCP with Cloud Run and Pub/Sub, processing **1M+ rows** with zero timeouts under burst concurrency.
- Devised a LangGraph and Gemini Map-Reduce ETL state machine with adaptive batching, achieving **99.8% schema validation accuracy** via programmatic retry transitions.
- Integrated DuckDB for vectorized in-memory aggregation, cutting compilation latency by **40%** while streaming real-time status updates in **under 200ms** via Firestore.

Quarantine: Fault-Tolerant Dead-Letter Queue | Python, AWS, GCP, Azure

GitHub | PyPI Package | 2026

- Authored an open-source fault-tolerant Python library implementing dead-letter queues (DLQ), guaranteeing non-blocking execution across batch loops of **10,000+ records**.
- Constructed cross-cloud storage abstractions across AWS S3, GCP, and Azure Blob to quarantine corrupted records, slashing data recovery time by **40%**.
- Orchestrated multi-platform distribution via GitHub Actions CI/CD to **PyPI** and **Conda-Forge**, maintaining semantic versioning and **over 90% test coverage**.

HONORS & ACHIEVEMENTS

Hack4Bengal 3.0 | 1st Prize Winner & Team Lead (3,000+ Applicants, 100+ Teams)

Jun 2024

- Won **1st Place out of 3,000+ applicants** as Team Lead of 3 engineers, architecting **PAWsitve** with Gemini Vision triage and emergency routing in **under 3 seconds**.

Competitive Programming | LeetCode Knight (Peak Rating: 1868, Top 6% Globally)

LeetCode Profile

- Solved **630+** algorithmic problems across graph theory, trees, concurrency, and DP, ranking in the top 6% of competitive programmers worldwide.

Research Publication | First Author, International Journal of Innovative Research in Technology

IJIRT Paper (View)

- Introduced a lightweight vision interaction engine achieving **97.3% gesture accuracy at 22ms latency**, reducing device power consumption by **41%** over ResNet-50.

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer and Electronics Engineering | CGPA: 8.53 / 10.0

Chennai, India

Sep 2021 -- Jul 2025